ATHIRA ARUN

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SUMMARY

B.Tech Computer Science (AI & ML) undergraduate passionate building intelligent systems using Artificial Intelligence, Natural Language Processing, and Machine Learning. Experienced in developing AI-driven applications including facial recognition systems, recommendation engines, and NLP-based analytics tools. Strong interest in conversational AI, deep learning, and scalable system design to solve real-world problems.

TECHNICAL SKILLS:

Programming Languages
Python, Java, C, SQL, JavaScript

Artificial Intelligence / Machine Learning
Natural Language Processing (NLP)
Large Language Models (LLMs)
Generative AI Systems
Emotion Detection (Deep Learning)
Recommendation & Matching Systems
Explainable AI (XAI)

Web & Backend Development
Flask
FastAPI
API Development
AI Model Integration
Database Integration

Tools & Technologies
Git & GitHub
OpenCV
TensorFlow / PyTorch
Arduino & IoT Sensors
HTML, CSS

Soft Skills
Problem Solving
Analytical Thinking
Project Documentation
Communication & Collaboration

INTERNSHIP EXPERIENCE

1. CYNARIS SOLUTIONS (Offline Internship Program) BANGALORE, KARNTAKA /INDIA
AI Intern | ERP Development Intern

05/06/2025 – 05-07-2025

Project1: Career Guidance Chatbot | Python, NLP, Streamlit, Generative AI

• Developed an AI-powered career guidance chatbot using Python, NLP, and Streamlit that analyzes user skills, interests, and education to generate personalized career recommendations.
• Integrated a large language model API to process structured prompts and produce detailed career insights, improving the accuracy and relevance of recommendations.
• Designed an interactive Streamlit interface enabling users to input preferences and receive AI-generated career suggestions within seconds.
• Implemented session history functionality to store and display previous queries, improving user interaction efficiency and accessibility of results.
• Optimized prompt engineering and response formatting to generate structured career summaries with learning resources and guidance.

01/06/2026 - 01/8/2026

Project2: ATS Enhancement using Frappe HRMS | Python, Frappe Framework, ERPNext, MariaDB, JavaScript

• Enhanced the Applicant Tracking System (ATS) by customizing recruitment workflows and candidate management features using the Frappe Framework and ERPNext.
• Developed backend functionality in Python for applicant processing, resume management, and recruitment automation.
• Customized DocTypes, forms, and workflows to streamline interview scheduling and candidate tracking.
• Implemented and debugged resume upload functionality by resolving issues related to file storage, database connectivity, and server configuration.
• Configured and managed MariaDB, Redis, and Bench while troubleshooting deployment and authentication issues on Ubuntu Linux.
• Developed client-side customizations using JavaScript to improve the recruitment workflow and user experience.

2. UNIFIED MENTOR (Online Internship Program)
Full stack web development intern *01/06/2025- 01/07/2025*

• Completed a structured internship focused on full stack web development covering frontend and backend concepts.
• Built web application components using HTML, CSS, JavaScript, and backend integration techniques.
• Learned REST API integration, database connectivity, and deployment workflows for web applications.

3. CODTECH IT SOLUTIONS (Online Internship Program)
Java programming intern *05/06/2025-05/06/2025*

• Developed applications using core Java concepts including object-oriented programming, classes, and data structures.
• Implemented modular programs focusing on problem solving, debugging, and clean code practices.
• Strengthened understanding of Java programming fundamentals and software development workflows.

4. MINDENIOUS EDUTECH (Online Internship/Training Program)
Machine learning intern

05/06/2026-04/08/2026

• Developed a real-time Credit Card Fraud Detection AI System that analyzes transactions using a machine learning model and business rules to identify fraudulent activities
• Built a banking-style fraud operations dashboard with live risk scores, alerts, customer profiles, analytics, and explainable AI (SHAP) for transparent decision-making.
• Implemented the project using React, FastAPI, Scikit-learn, PostgreSQL/SQLite, and JWT authentication, with secure multi-user access and real-time transaction monitoring.

PROJECTS

Auto-Activated Defensive Device Triggered by Force & Stress Detection

• Developed an IoT-based personal safety system using ESP32, integrating heart rate and temperature sensors to detect stress and potentially dangerous situations in real time.
• Implemented automated alert mechanisms, SOS emergency triggering, LCD-based monitoring, and Blynk-powered remote notifications for enhanced personal security and emergency response.

ATS Enhancement using Frappe HRMS

• Developed a customized Applicant Tracking System (ATS) using Frappe Framework and ERPNext HRMS to automate candidate tracking, resume management, and interview workflows.
• Implemented Python-based backend customizations, configured MariaDB and Redis, and enhanced recruitment processes through custom DocTypes and workflow automation.

Credit Card Fraud Detection AI System

• Developed a real-time AI-powered Credit Card Fraud Detection System using Machine Learning, FastAPI, and React to identify fraudulent transactions through risk-based predictions and business rules.
• Implemented a secure full-stack platform with JWT authentication, transaction monitoring, customer profiles, alerts, and analytics using PostgreSQL/SQLite for fraud management.
• Integrated Explainable AI (SHAP) and an interactive fraud operations dashboard to provide transparent predictions, risk explanations, and actionable insights for financial security.

Real-Time Invisibility Cloak

• Developed a real-time computer vision application using Python, OpenCV, and NumPy that makes a red cloth disappear through webcam-based background replacement and color segmentation.
• Implemented HSV color detection, image processing, and temporal smoothing to create a smooth invisibility effect with real-time performance and interactive controls.

VidyaAI Tutor – AI-Powered Educational Assistant

• Developed an AI tutoring platform that delivers curriculum-based answers using semantic retrieval and context pruning, reducing token usage and operational costs by up to 85%.
• Built a scalable React and TypeScript application with intelligent content retrieval, enabling fast, low-bandwidth educational support for students in resource-constrained environments.

AI-Powered Attendance System

• Developed a facial recognition-based attendance system using computer vision.
• Implemented automated attendance tracking with report generation and anti-spoofing security features.

Colorizer AI – Deep Learning Image Colorization System

• Developed an AI-powered image colorization application using Deep Learning, OpenCV, and Streamlit to automatically convert grayscale images into realistic colorized outputs.
• Implemented image enhancement features including brightness adjustment, contrast improvement, sharpening, and before-after comparison with downloadable results.

Neural Student Predictor – Neural Network Performance Prediction System

• Developed a neural network from scratch using Python and NumPy, implementing forward propagation, backpropagation, gradient descent, and cross-entropy loss without deep learning frameworks.
• Built an interactive Streamlit application that predicts student performance levels based on academic metrics and visualizes model training through loss curve analysis.

AI Resume–Job Matching System

• Built an intelligent system that performs semantic matching between resumes and job descriptions.
• Implemented skill gap analysis using NLP techniques to recommend improvements.

Smart Interview Preparation AI

• Developed an AI tool that generates role-specific interview questions using NLP techniques.
• Implemented automated evaluation logic to help candidates assess their preparation level.

Fake News Detection System

• Created a machine learning model for classifying news articles as real or fake.
• Implemented explainable AI techniques to provide transparency in model predictions.

AI-Based Personal Finance Advisor

• Developed a machine learning system for analyzing expenses and predicting savings patterns.
• Applied time-series analysis for financial trend prediction.

Emotion Detection System

• Built a deep learning model using CNN architectures to detect facial emotions in real time.
• Integrated computer vision techniques for image processing and classification.

Career Guidance Chatbot

• Developed an NLP-based chatbot that provides personalized career recommendations.
• Implemented conversational AI techniques to improve interaction and decision support.

TaskFlow Manager – MERN Stack Task Management Application

• Developed a full-stack task management web application using the MERN stack with JWT-based authentication, user-specific task isolation, and secure CRUD operations.
• Implemented advanced task management features including search, filtering, sorting, priority tracking, status management, and a responsive React-based dashboard.

ShopSphere – MERN Stack E-Commerce Web Application

• Developed a full-stack e-commerce platform using the MERN stack, featuring secure JWT authentication, RESTful APIs, and MongoDB database integration.
• Designed a responsive React-based user interface and scalable backend architecture to support product management, shopping cart functionality, and order processing.

EDUCATION

SCHOOLING

Silicon City Academy Of Secondary Education *Year of study:2012-2024*

Bangalore, Karnataka/India

UNIVERSITY

Karunya Institute of Technology and Sciences *Year of study:2024-2028*

Coimbatore, TamilNadu/ India

LANGUAGES:

• English — Fluent
• Kannada — Fluent
• Malayalam — Native/Fluent
• Tamil — Fluent
• Hindi — Basic
• Telugu — Basic

OTHER EDUCATION:

- Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate — Oracle University
- Artificial Intelligence Fundamentals — IBM SkillsBuild
- Data Science — Infosys Springboard
- Programming Fundamentals using Python — Infosys Springboard
- Java Programming Fundamentals — Infosys Springboard
- Introduction to Large Language Models (LLMs) — NPTEL (Elite), IIT Madras
- Google Cloud Computing — NPTEL (Elite)
- Software Conceptual Design — NPTEL (Elite)
- Red Hat System Administration I (RH124 – RHA) — Red Hat

· Cisco Certified Network Associate (CCNA) Program: Introduction to Networks, Switching, Routing & Wireless Essentials, and Enterprise Networking, Security & Automation — Cisco Networking Academy

OTHER INTERESTS:

• Dance
• Acting